# Housing Prices Analysis Report

## 1. Data Cleaning and Preparation

The data was initially loaded from 'train.csv' and 'test.csv'. Missing values in numeric columns were filled using the mean of each column. Columns with a high percentage of missing values ('Alley', 'PoolQC', 'MiscFeature') were dropped to simplify the dataset. The cleaned data was then saved to 'cleaned_data.csv'.

## 2. Predictive Model Development

A RandomForestRegressor was used to develop the predictive model. The model was trained on 80% of the data with 'SalePrice' as the target variable. The remaining 20% was used for testing. Model parameters included 100 trees with a random state set for reproducibility. The predicted prices were saved in 'predicted_prices.csv'.

## 3. Model Evaluation

The model's performance was evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics provide insights into the average error in predictions and the standard deviation of these errors, respectively. The results were saved in 'model_evaluation_metrics.csv'.

## 4. Data Visualization

Various visualizations were created to explore the data further. These include histograms for 'LotArea' and 'YearBuilt', box plots for 'LotFrontage' and 'OverallQual', and scatter plots showing the relationship between 'GrLivArea' and 'SalePrice'. A count plot for 'Neighborhood' was also generated. All visualizations were saved in 'visualizations.png'.

## 5. Cluster Analysis

# Housing Prices Analysis Report

A cluster analysis was performed using KMeans clustering on the principal components of the numeric data. This analysis helps in identifying patterns or groups within the data. The results were visualized and saved in 'cluster_analysis.png'.

# Housing Prices Analysis Report

## Visualizations